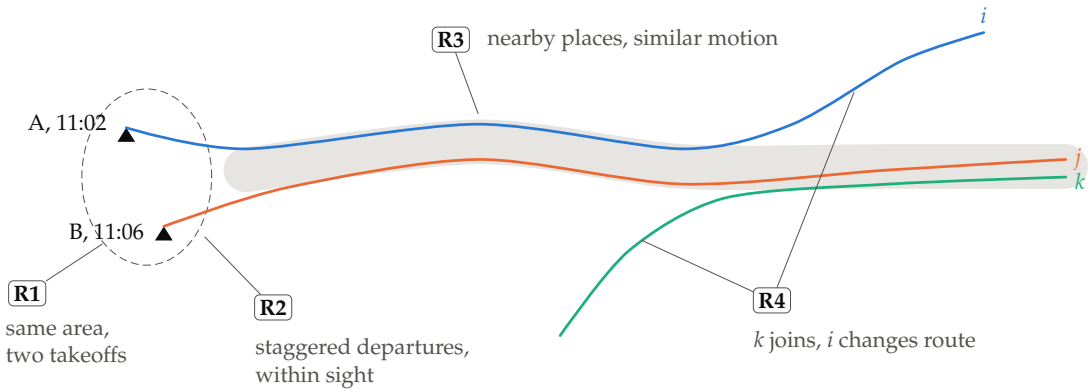
Identifying group flights in the FFVL archive

Part 1: the idea

What the method does, on pictures

The question

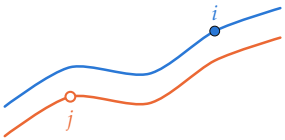
For every flight and every instant: alone or in a group, and with whom?



Pilots in a group need not take off from the same site, like *j* and *k*.

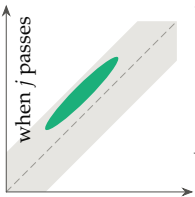
The method in six steps

1 Two tracks



j flies a similar route,
a little later

2 Same place, with a delay



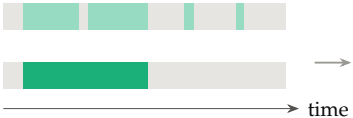
both pass
the same
place;
above the
diagonal:
j passes
later

3 Contact



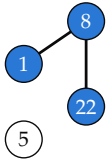
when one of the two pilots
is on the path of the other

4 Long contacts



short gaps filled,
short pieces dropped

5 Configurations, groups



14

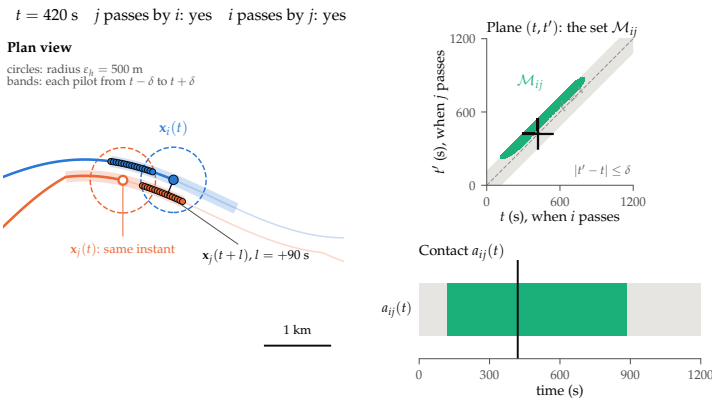
linked pilots:
configuration;
same members
in time: group

6 Episodes



groups that follow one
another, sharing a member

Two pilots on similar routes, 90 s apart

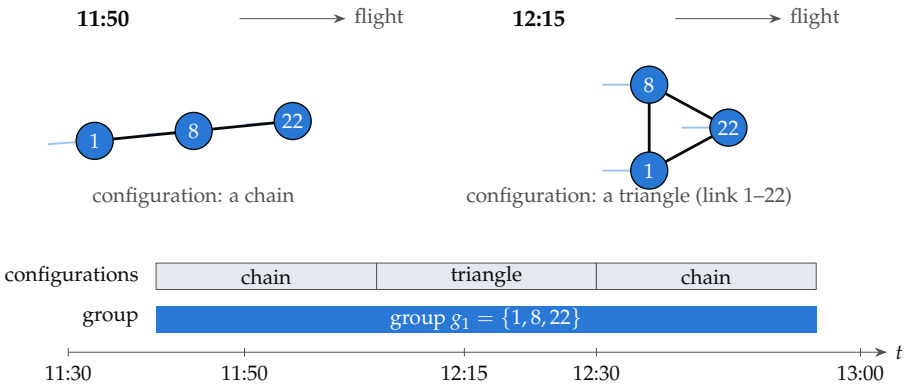


j flies a route similar to that of i, here 90 s later. In reality the delay varies: we want it within δ.

Same instant: here about 0.8 km apart, and it varies too.

Same place (within ε_h), with a delay (within δ), seen from i or from j: **in contact**.

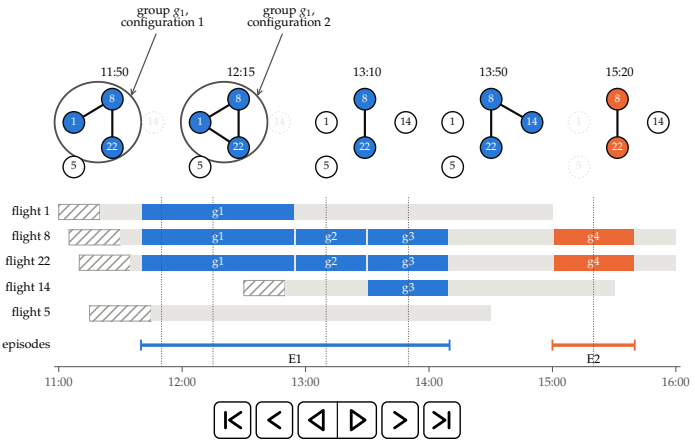
Configuration and group



Configuration: gliders linked by contacts, with their links. A link more or less: a new configuration.

Group: the same gliders, from a start to an end. It changes only when a glider joins or leaves.

From configurations to episodes: an example day



Configuration at t: pilots linked by chains of contacts, with their links.

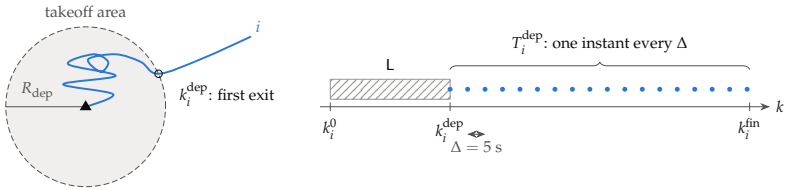
Group: the same members, from a start to an end. An arrival or a departure starts a new one ($g_1, \ldots, g_4$).

Episode: a chain of groups, each group starting the instant the previous group ends and sharing a member with it: $g_1 \to g_2 \to g_3$ is E1. g_4 starts 50 min after g_3 ends: a new episode, E2.

Part 2: quantitative overview

The same steps, with formulas

Start at the departure



Departure.

$$k_i^{\text{dep}} = \min\{k : \ \|\mathbf{x}_i(k) - \mathbf{x}_i(k_i^0)\| > R_{\text{dep}}\}, \qquad R_{\text{dep}} = 3 \text{ km}.$$

Before it: local phase L.

Evaluation instants.

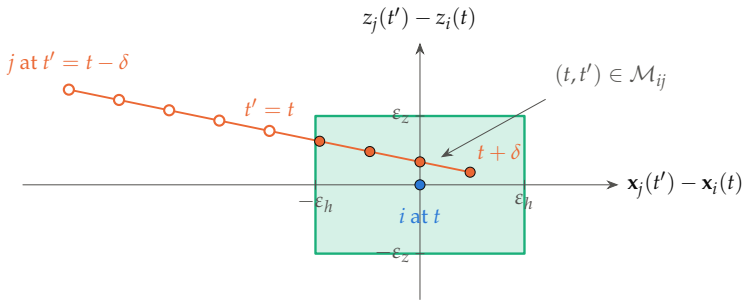
$$T_i^{\text{dep}} = \{t \in \Delta\mathbb{Z} : \ k_i^{\text{dep}} \leq t \leq k_i^{\text{fin}}, \ i \text{ has a position at } t\}, \qquad \Delta = 5 \text{ s}.$$

Correspondence: the same place at two instants

Correspondence. One clock (UTC) for all flights: i at instant t, j at instant $t' = t + l$, with l the delay:

$$(t, t') \in \mathcal{M}_{ij} \iff |t' - t| \leq \delta,\ \ \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| < \varepsilon_h,\ \ |z_i(t) - z_j(t')| < \varepsilon_z,\ \ D_{ij}(t, t') = 1$$

D_{ij} is the heading condition (next slide).

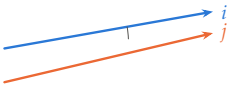


$\delta = 2$ min
$\varepsilon_h = 500$ m
$\varepsilon_z = 300$ m

Heading condition: ruling out chance crossings

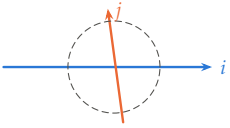
Pilots who cross on different routes pass the distance and delay tests by chance. A group flies the same way.

(a) both glide



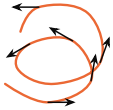
angle $< \theta_{\max}$: ✓

(b) crossing



close, heading elsewhere: ×, excluded

(c) climb or search

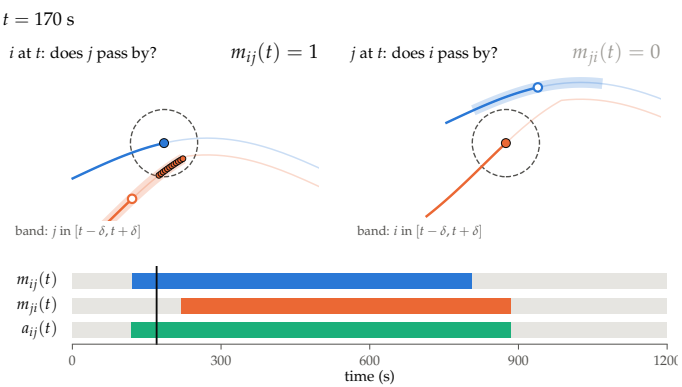


the heading turns: no condition

Heading condition.

$$D_{ij}(t,t') = \begin{cases} \mathbf{1}\left[\hat{\mathbf{v}}_i(t) \cdot \hat{\mathbf{v}}_j(t') > \cos\theta_{\max}\right] & \text{if both pilots glide,} \\ 1 & \text{otherwise,} \end{cases} \qquad \theta_{\max} = 45^\circ.$$

Contact: what i sees, what j sees



Contact.

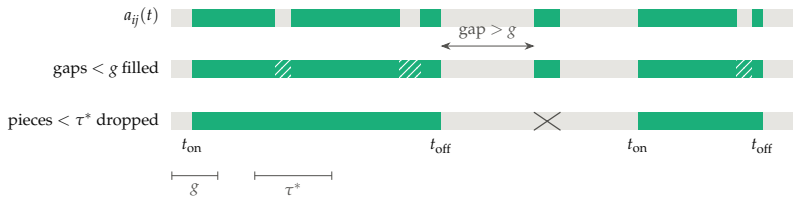
$$m_{ij}(t) = \mathbf{1}\left[\exists\, t' : (t, t') \in \mathcal{M}_{ij}\right]$$
$$m_{ji}(t) = \mathbf{1}\left[\exists\, t' : (t', t) \in \mathcal{M}_{ij}\right]$$
$$a_{ij} = m_{ij} \vee m_{ji}$$

j joins the route of i, 90 s behind, then leaves it.

Start: $m_{ij} = 1$, $m_{ji} = 0$. End: the reverse.

$m_{ij} \neq m_{ji}$ in general; the contact $a_{ij} = a_{ji}$ is symmetric.

Pair intervals: fill short gaps, drop short contacts



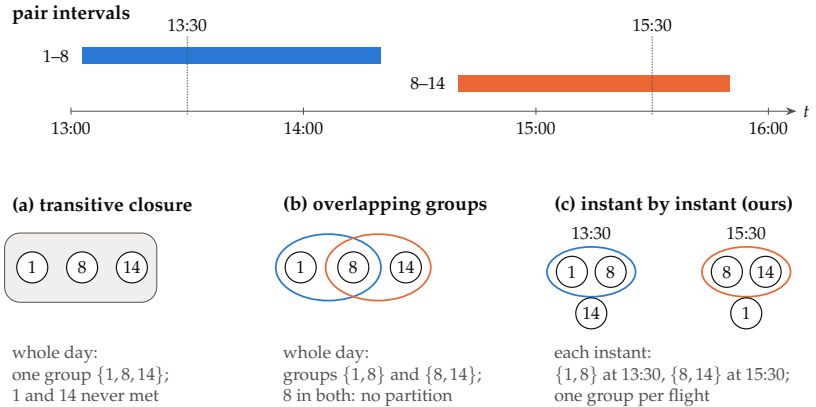
Pair interval. A run of $a_{ij} = 1$, gaps $< \boldsymbol{g = 120}$ **s** joined, lasting $\geq \boldsymbol{\tau^* = 38.1}$ **min**: $[t_{\text{on}}, t_{\text{off}})$.

Choosing τ^*. Not set by hand: we fix $q = 10\%$, the share of kept intervals we accept to be chance. The null model (pairs shifted in time) counts the chance intervals $C(\tau)$ among the $N(\tau)$ observed:

$$\tau^* = \min\left\{\tau : \frac{C(\tau)}{N(\tau)} \leq q\right\}.$$

Smaller τ^*: more intervals, more of them chance. Larger τ^*: fewer, cleaner.

Contact is not transitive: three ways to define groups



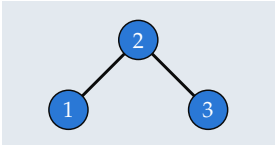
(a) and (b): one picture of the whole day, without time. (c): one picture per instant; 8 is in two groups, never at the same instant.

Active graph: the configurations at one instant

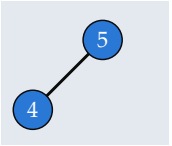
Active graph.

$$G(t) = \big(V(t), E(t)\big), \qquad V(t) = \text{flights in the air at } t, \qquad E(t) = \big\{\{i,j\} : \ t \in [t_{\text{on}}, t_{\text{off}})\big\}.$$

$G(t)$ **at one instant** t



configuration
a chain 1–2–3



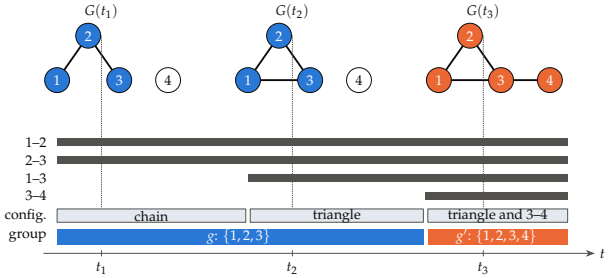
configuration
$\{4,5\}$



solo

Configuration. At t: a connected component of $G(t)$ with two or more flights, with its edges.

Group: fixed members, changing configurations



Group. $g = (M_g, [t_g^{\text{start}}, t_g^{\text{end}}))$: M_g are the members of a configuration of $G(t)$ at every t of the interval, and the interval is maximal.

Membership.

$$\gamma_i(t) = \begin{cases} \mathsf{L} & \text{before the departure,} \\ g & i \in M_g,\ t \in [t_g^{\text{start}}, t_g^{\text{end}}), \\ \mathsf{S} & \text{otherwise (solo).} \end{cases}$$

Episode: groups linked in time



Episode. Link $g \to g'$: g' starts the instant g ends, and they share a member,

$$t^{\text{start}}_{g'} = t^{\text{end}}_{g}, \qquad M_g \cap M_{g'} \neq \varnothing.$$

An episode: a connected component of the graph of these links. g_1 and g_2 have no link between them, but both link to g_3: all three are in E1.

On the FFVL archive

21.7%
of paraglider flights fly in a group at some point

6.9%
of hang-glider flights

17 169
episodes

58 min
median episode; 2 flights (median), up to 53

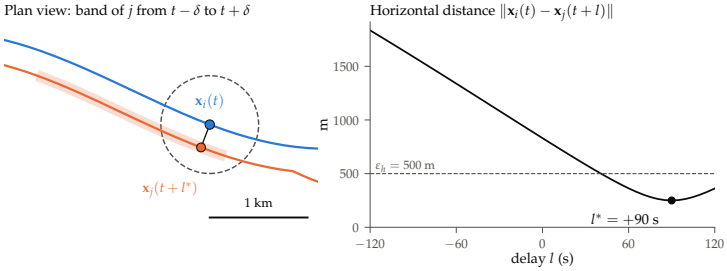
160 637 flights; $\delta = 2$ min, $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m, $g = 120$ s, $\tau^* = 38.1$ min.

Who follows whom: the delay in a glide

The contact needs only that some delay within δ exists; its sign says who follows whom.

Delay (Li et al.). a label, not used to build the groups. At t, where j passes closest to i; over a pair interval, the median:

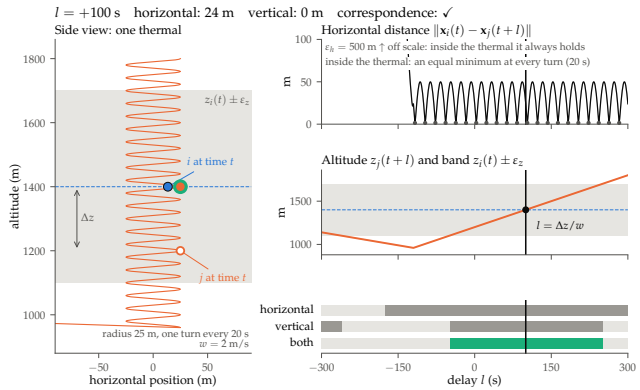
$$l^*(t) = \arg\min_{l:\,(t,t+l)\in\mathcal{M}_{ij}} \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|, \qquad \ell_{\text{med}} = \text{median}\, l^*, \qquad \ell_{\text{med}} > 0 :\ j \text{ follows } i.$$



Glide: a nearly straight path, so the distance has one sharp minimum: l^* is the delay at which j crosses the point of i.

Model: j on the route of i, 90 s behind and 250 m to the side, at 8 m/s; $\delta = 2$ min.

Who follows whom: the delay in a thermal, not in search



Thermal: j circles: the distance has an equal minimum at every turn.

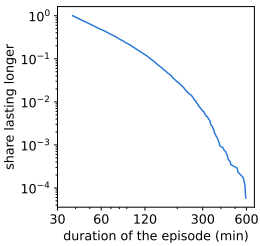
Altitude narrows l:
$z_j(t+l) \simeq z_j(t) + w\,l$ climbing at w, so with $\Delta z = z_i(t) - z_j(t)$

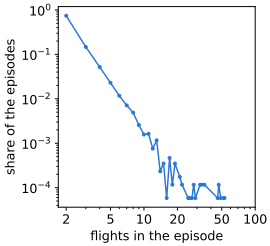
$$l \in \left(\frac{\Delta z - \varepsilon_z}{w},\ \frac{\Delta z + \varepsilon_z}{w} \right),$$

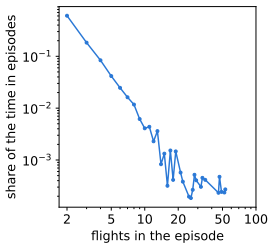
around $\Delta z / w$, the time j needs to reach i. Here $(-50, 250)$ s, cut by δ to $(-50, 120]$ s.

Search: $w \simeq 0$, one minimum per turn and no altitude window: left out of ℓ_{med}.

Episodes: how long, how many flights







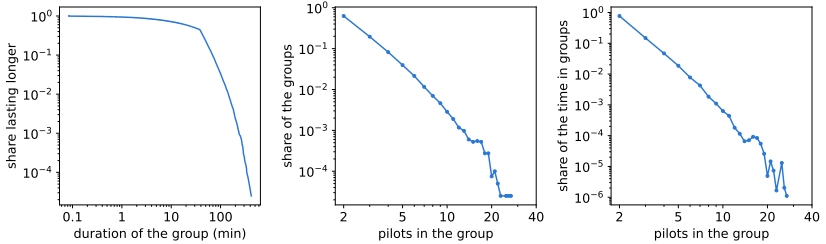
$$y = \frac{\#\{E:\ D_E \geq d\}}{N} \qquad y = \frac{N_n}{N} \qquad y = T(n) = \frac{\sum_{E:\, n_E = n} D_E}{\sum_E D_E}$$

Episode E: D_E from the start of its first group to the end of its last, n_E its distinct flights. $N = 17\,169$ episodes, N_n of them with $n_E = n$. Whole archive, $\delta = 2$ min, $q = 0.1$.

$T(n)$: of every 100 s of episode time, summed over all episodes, on average $100\,T(n)$ s are in episodes of n flights; for $n = 5$, 4.2 s.

With $q = 0.1$ the median episode lasts 58 min and 74% are pairs. Weighted by time the pairs drop to 61%: large episodes last longer.

Groups: how long, how many pilots at once



$$y = \frac{\#\{g:\ D_g \geq d\}}{N} \qquad y = \frac{N_n}{N} \qquad y = T(n) = \frac{\sum_{g:\, n_g = n} D_g}{\sum_g D_g}$$

Group g: $D_g = t_g^{\text{end}} - t_g^{\text{start}}$, $n_g = |M_g|$ its pilots; each arrival or departure starts a new group. $N = 40\,051$ groups, N_n of them with $n_g = n$. Whole archive, $\delta = 2$ min, $q = 0.1$.

$T(n)$: of every 100 s of group time, summed over all groups, on average $100\,T(n)$ s are in groups of n pilots; for $n = 5$, 1.9 s. A group counts its duration once, not once per pilot.

Median group 30 min, 63% are pairs. Weighted by time the pairs rise to 77%: large groups change quickly.